Chapter test

6 Food and humans

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. Which of the following best describes a heterotroph?

   A. It is able to synthesize organic molecules from inorganic substances.
   B. It obtains nutrients from living organisms only.
   C. It lives on dead organic matter.
   D. It obtains organic nutrients from the environment. ☐

★ 2. The table below shows the nutritional compositions of four foods.

Food	Carbohydrates (%)	Lipids (%)	Proteins (%)	Water (%)
P	3	14	26	57
Q	10	65	12	13
R	7	6	9	78
S	22	17	26	35

Which food can provide the most energy per gram?

A. P
B. Q
C. R
D. S ☐

3. Excess glucose will be stored in the human body as

(1) glycogen.
(2) starch.
(3) fat.

A. (1) and (2) only
B. (2) and (3) only
C. (1) and (3) only
D. (1), (2) and (3) ☐

© 2019 Aristo Educational Press Ltd.

★ 4. It is suggested that when babies get older, they should not be fed with milk only. Babies fed only with milk for a long period of time may develop anaemia because

A. milk contains a lot of lactose.
B. milk is rich in proteins.
C. milk does not contain any roughage.
D. milk has a low content of iron. ☐

★ 5. A student carried out various tests on a food sample. The table below shows the results.

	Iodine test	Benedict's test	Grease spot test
Result	A brown solution resulted.	Brick-red precipitates formed.	The spot on the filter paper disappeared after drying.

The food sample contains

A. sucrose.
B. starch.
C. glucose.
D. lipids. ☐

Direction: Questions 6 and 7 refer to the nutrition label shown on the right.

★ 6. Which of the following is most likely the food product bearing this nutrition label?

A. Wheat bread
B. Tofu
C. Coke
D. Sausages ☐

★ 7. If a teenager eats this food product only for a month, which of the following conditions may appear?

A. Diarrhoea
B. Constipation
C. Anaemia
D. Rickets ☐

Amount Per Serving

Calories 280 Fat Cal 130

	% Daily Value*
Total Fat 14g	**22%**
Saturated Fat 1.5g	**8%**
Trans Fat 0g	
Cholesterol 0mg	**0%**
Sodium 620mg	**25%**
Potassium 260mg	**7%**
Total Carbohydrate 8g	**3%**
Dietary Fiber 1g	**4%**
Sugars 3g	
Protein 30g	

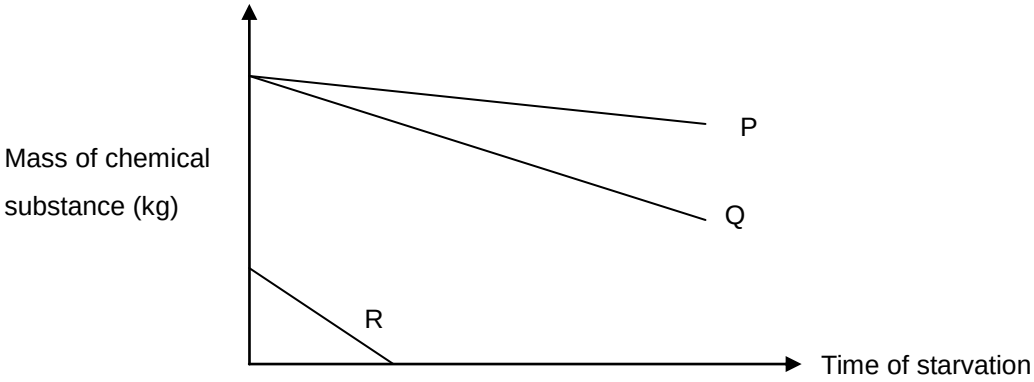
Vitamin A 2% • Vitamin C 0%
Calcium 4% • Iron 20%

* Percent Daily Values are based on a 2,000 calorie diet. Your daily values may be higher or lower depending on your calorie needs:

	Calories	2,000	2,500
Total Fat	Less than	65g	80g
Sat Fat	Less than	20g	25g
Cholesterol	Less than	300mg	300mg
Sodium	Less than	2,400mg	2,400mg
Total Carbohydrate		300g	375g
Dietary Fiber		25g	30g

© 2019 Aristo Educational Press Ltd.

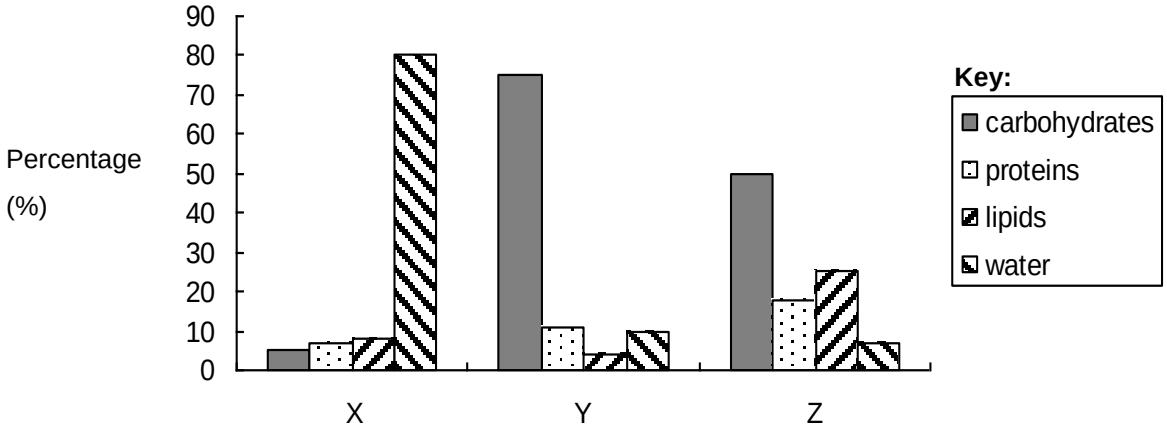
★★ 8. The graph below shows the changes in the mass of three chemical substances in the human body during a month of starvation.



What do P, Q and R represent respectively?

	P	***Q***	***R***
A.	proteins	lipids	glycogen
B.	glycogen	lipids	proteins
C.	lipids	proteins	glycogen
D.	proteins	glycogen	lipids

Direction: Questions 9 and 10 refer of the graph below, which shows the percentage of carbohydrates, proteins, lipids and water in three foods, X, Y and Z.



★★ 9. Which of the following most likely identifies foods X, Y and Z?

	X	***Y***	***Z***
A.	rice	peanut	milk
B.	milk	peanut	rice
C.	milk	rice	peanut
D.	rice	milk	peanut

© 2019 Aristo Educational Press Ltd.

★★ 10. Which of the following correctly arranges the energy content of per unit mass of foods X, Y and Z?

A. X > Y > Z
B. X > Z > Y
C. Z > X > Y
D. Z > Y > X

Question	1	2	3	4	5	6	7	8	9	10
Related section	6.1	6.2	6.2	6.2	6.3	6.2	6.2	6.4	6.2	6.2

B. Problem solving (19 marks)

★ 11. Table 1 shows the nutrition information of bananas and avocados.

Nutrient	Banana (per 100 g)	Avocado (per 100 g)
Fat	0.2 g	15.3 g
Cholesterol	0.0 g	0.0 g
Carbohydrate	22.0 g	7.4 g
Fibre	1.2 g	2.1 g
Protein	1.4 g	2.00 g
Calcium	7.0 mg	11.0 mg
Iron	0.4 mg	1.0 mg
Vitamin C	8.0 mg	8.0 mg

Table 1

Table 2 shows the recommended daily intake for calcium, iron and vitamin C.

Nutrient	Recommended daily intake
Calcium	550 mg
Iron	8.7 mg
Vitamin C	30 mg

Table 2

(a) Although bananas can support daily activities, children are not recommended to have a diet containing bananas only. Explain why. (3 marks)

© 2019 Aristo Educational Press Ltd.

(b) (i) Which fruit contains a higher energy value? (1 mark)

(ii) Compare the nutrient content of bananas and avocados. Explain why many runners eat bananas, instead of eating avocados, just before a running race. (3 marks)

(Total: 7 marks)

★ 12. Mary performed various food tests to investigate the chemical compositions of three foods (A, B and C). She then repeated the food tests with the extracts of A, B and C which had been incubated with amylase and lipase for 15 minutes. The results are summarized in the table below.

	Without enzymes			**After incubated with enzymes**			
	Iodine test	**Benedict's test**	**Grease spot test**	**Iodine test**	**Benedict's test**	**Grease spot test**	**Change in pH**
A	Blue black	Blue	Translucent spot left	Brown	Orange	Translucent spot left	Decreased
B	Brown	Orange	Translucent spot left	Brown	Orange	No translucent spot left	Decreased
C	Brown	Blue	No translucent spot left	Brown	Blue	No translucent spot left	Remain unchanged

(a) A, B and C are all solid foods. Describe how Mary would treat the foods before carrying out the food tests. (3 marks)

© 2019 Aristo Educational Press Ltd.

(b) Based on the results, state the type of carbohydrate that each food contains. (3 marks)

A: _____

B: _____

C: _____

(c) (i) Explain why the pH value of the extracts of A and B decreased after incubating with enzymes. (2 marks)

_____

_____

_____

(ii) Which food has the highest lipid content? Explain why. (4 marks)

_____

_____

_____

_____

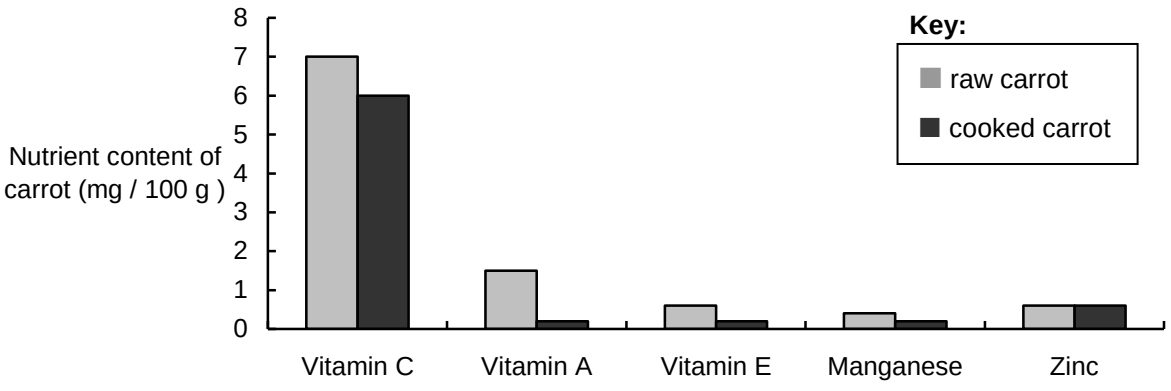
_____

_____

(Total: 12 marks)

C. Exam challenge (11 marks)

★★ 13. Some nutrients are heat-sensitive. They may be lost during the cooking process. The graph below compares the content of some major nutrients in raw carrots and boiled carrots.



© 2019 Aristo Educational Press Ltd.

(a) Based on the graph, describe how boiling affects the nutrient content of carrots. (3 marks)

(b) A student suggested that less vitamin C would be lost when carrots are cooked using a microwave oven instead of boiling with water. Design an investigation to test his hypothesis. (5 marks)

(c) Predict and explain the results of the investigation you designed in (b). (3 marks)

(Total: 11 marks)

© 2019 Aristo Educational Press Ltd.

Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	D	2.	B	3.	C	4.	D	5.	C
6.	D	7.	B	8.	A	9.	C	10.	D

(1 mark each)

B. Problem solving (19 marks)

11. (a) The protein content of bananas is too low to provide enough proteins for the growth and repair of tissues. (1)
The calcium content of bananas is too low for healthy bone and tooth development. (1)
The iron content of bananas is too low for producing enough red blood cells / haemoglobin. (1)

(b) (i) Avocado (1)

(ii) Bananas have a higher carbohydrate content than avocados. / Avocados have a higher fat content than bananas. (1)
Carbohydrates can be easily digested to glucose. The digested products of fats have to undergo some chemical reactions in order to be converted to glucose. (1)
Therefore, compared with avocados, bananas can provide more glucose molecules for cells to carry out respiration which provides energy for muscle contraction within a shorter period of time. (1)

(Total: 7 marks)

12. (a) Grind the food with distilled water. (1)
Filter the mixture. (1)
Collect the filtrate to conduct the food tests. (1)

(b) A: starch (1)
B: reducing sugars (1)
C: sucrose / cellulose (1)

(c) (i) Lipids in A and B were broken down by lipase (1)
to form fatty acids, which lower the pH of the food extracts. (1)

(ii) A has the highest lipid content. (1)
Only A and B contain lipids since translucent spots were left in the grease spot test. (1)
For B, after incubating it with lipase, there was no translucent spot left in the grease spot test, which means all lipids were broken down within 15 minutes. (1)

© 2019 Aristo Educational Press Ltd.

While there was still translucent spot left for A, this suggested that A contained more lipids than B because the breakdown of lipids was not completed within the same period of time. (1)

(Total: 12 marks)

C. Exam challenge (11 marks)

13. (a) Any ***three*** of the following: (3)
    - All vitamin contents decrease after boiling. (1)
    - Manganese content decreases after boiling. (1)
    - Boiling has no effect on zinc content. (1)
    - Boiling has the biggest effect on vitamin A content. (1)

    (b) Cook the same amount of carrots with a microwave oven and with boiling water respectively. (1)
    Extract juice from the carrots cooked by each method. (1)
    Titrate the juices with DCPIP solution. (1)
    Observe the colour change of the DCPIP solution and compare the volume of DCPIP solution used to reach the end point of titration. (1)
    Repeat the titration at least three times to obtain the average volume of DCPIP solution needed to reach the end point of titration. (1)

    (c) When carrots are cooked using a microwave oven, less vitamin C would be lost compared with boiling in water. (1)
    This is because vitamin C is water-soluble. (1)
    When cooked in boiling water, vitamin C in carrots would dissolve in water and be destroyed by heat. (1)

(Total: 11 marks)

© 2019 Aristo Educational Press Ltd.